

A Probabilistic Proof of the Johnson–Lindenstrauss Lemma

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1 Introduction

The Johnson–Lindenstrauss lemma [1] is a classical result in high-dimensional probability. It states that any finite set of points in a high-dimensional Euclidean space can be embedded into a space of much lower dimension while nearly preserving all pairwise distances. The lemma has found wide application in engineering fields such as coding theory, embedding theory, and generative modeling.

Lemma 1.1 (Johnson–Lindenstrauss). *Let $0 < \epsilon < 1$, let P be a set of $N \geq 2$ points in $\mathbb{R}^n$, and let m be an integer with*

$$m > \frac{16 \ln N}{\epsilon^2}.$$

Then there exists a linear map $M : \mathbb{R}^n \rightarrow \mathbb{R}^m$ such that

$$(1 - \epsilon)\|u - v\|^2 \leq \|Mu - Mv\|^2 \leq (1 + \epsilon)\|u - v\|^2 \quad \text{for all } u, v \in P.$$

Note that the guarantee holds for the finitely many points of P , not for all of $\mathbb{R}^n$: when $m < n$, any linear map has a nontrivial kernel, so no linear map can preserve the norms of *all* vectors in $\mathbb{R}^n$.

Remark. We make no attempt to optimize the constant 16 here. It is simply what the tail bound in Section 3 produces. For comparison, the statement of the lemma on Wikipedia [2] has the constant 8.

The proof presented here is probabilistic: we construct a random matrix and show that it has the desired property with positive probability. This argument is adapted from the 2026 Concours BCE mathematics exam [3]. I have summarized their approach and translated it into English.

2 A Random Matrix Construction

Fix a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ carrying independent and identically distributed standard Gaussian random variables $G_{i,j} : \Omega \rightarrow \mathbb{R}$ with $G_{i,j} \sim \mathcal{N}(0, 1)$, indexed by $(i, j) \in [m] \times [n]$. For each $\omega \in \Omega$ we form the random $m \times n$ matrix

$$M(\omega) = \frac{1}{\sqrt{m}} \begin{pmatrix} G_{1,1}(\omega) & \cdots & G_{1,n}(\omega) \\ \vdots & & \vdots \\ G_{m,1}(\omega) & \cdots & G_{m,n}(\omega) \end{pmatrix}.$$

Fix a vector $x \in \mathbb{R}^n$ with $\|x\| = 1$. (We use a lowercase letter to emphasize that x is a fixed, deterministic vector, not a random variable.) Define, for $i \in [m]$,

$$Y_i = \sum_{j=1}^n x_j G_{i,j}, \quad \text{so that} \quad Z := \|Mx\|^2 = \frac{1}{m} \sum_{i=1}^m Y_i^2.$$

Each summand $x_j G_{i,j}$ is a scaled Gaussian, distributed as $\mathcal{N}(0, x_j^2)$ (where $\mathcal{N}(0, 0)$ denotes the point mass at 0). Since the $G_{i,j}$ are independent, Y_i is a sum of independent Gaussians and is therefore itself Gaussian:

$$Y_i \sim \mathcal{N}\left(0, \sum_{j=1}^n x_j^2\right) = \mathcal{N}(0, \|x\|^2) = \mathcal{N}(0, 1).$$

Moreover, $Y_1, \dots, Y_m$ are independent, since they are built from disjoint rows of Gaussians. In particular,

$$\mathbb{E}[Y_i^2] = \text{Var}(Y_i) + \mathbb{E}[Y_i]^2 = \|x\|^2 = 1, \quad \text{hence} \quad \mathbb{E}[Z] = 1.$$

The heart of the proof is to show that Z concentrates sharply around this mean.

3 Concentration of $\|Mx\|^2$

We first compute the moment generating function of Y_i^2 , the square of a standard Gaussian.

Lemma 3.1. *If $Y \sim \mathcal{N}(0, 1)$, then for every $t < \frac{1}{2}$,*

$$\mathbb{E}[e^{tY^2}] = \frac{1}{\sqrt{1-2t}}.$$

Proof. Substituting $u = \sqrt{1-2t}x$ in the last step,

$$\mathbb{E}[e^{tY^2}] = \int_{-\infty}^{\infty} e^{tx^2} \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dx = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-(1-2t)\frac{x^2}{2}} dx = \frac{1}{\sqrt{1-2t}} \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{u^2}{2}} du = \frac{1}{\sqrt{1-2t}}. \quad \square$$

Proposition 3.2. *Let $0 < \epsilon < 1$ and let Z be as above. Then*

$$\mathbb{P}(|Z - 1| > \epsilon) \leq 2e^{-\frac{m\epsilon^2}{8}}.$$

Proof. Upper tail. Fix $t \in (0, \frac{1}{4}]$. Since $s \mapsto e^{tms}$ is strictly increasing, the events $\{Z \geq 1 + \epsilon\}$ and $\{e^{tmZ} \geq e^{tm(1+\epsilon)}\}$ coincide, so Markov's inequality gives

$$\mathbb{P}(Z \geq 1 + \epsilon) = \mathbb{P}(e^{tmZ} \geq e^{tm(1+\epsilon)}) \leq \mathbb{E}[e^{tmZ}] e^{-tm(1+\epsilon)}.$$

Since $mZ = \sum_{i=1}^m Y_i^2$ with the Y_i independent, the moment generating function factorizes, and Lemma 3.1 yields

$$\mathbb{E}[e^{tmZ}] = \prod_{i=1}^m \mathbb{E}[e^{tY_i^2}] = \left(\frac{1}{\sqrt{1-2t}} \right)^m.$$

Combining the two displays,

$$\mathbb{P}(Z \geq 1 + \epsilon) \leq \left(\frac{e^{-t}}{\sqrt{1-2t}} \right)^m e^{-tm\epsilon}.$$

We now use the elementary inequality

$$-t - \frac{1}{2} \ln(1 - 2t) \leq 2t^2 \quad \text{for } t \in [0, \frac{1}{4}], \quad (1)$$

which follows by noting that $g(t) = 2t^2 + t + \frac{1}{2} \ln(1 - 2t)$ satisfies $g(0) = 0$ and $g'(t) = \frac{2t(1-4t)}{1-2t} \geq 0$ on $[0, \frac{1}{4}]$. Inequality (1) gives $\frac{e^{-t}}{\sqrt{1-2t}} \leq e^{2t^2}$, hence

$$\mathbb{P}(Z \geq 1 + \epsilon) \leq e^{m(2t^2 - t\epsilon)}.$$

This holds for every $t \in (0, \frac{1}{4}]$, so we may choose the minimizer $t = \frac{\epsilon}{4} \in (0, \frac{1}{4})$ of $2t^2 - t\epsilon$, which yields

$$\mathbb{P}(Z \geq 1 + \epsilon) \leq e^{-\frac{m\epsilon^2}{8}}.$$

Lower tail. The argument is symmetric. For $t > 0$, applying Markov's inequality to e^{-tmZ} and using $\mathbb{E}[e^{-tY_i^2}] = \frac{1}{\sqrt{1+2t}}$ (Lemma 3.1 with $-t$ in place of t),

$$\mathbb{P}(Z \leq 1 - \epsilon) \leq \left(\frac{e^t}{\sqrt{1+2t}} \right)^m e^{-tm\epsilon} \leq e^{m(t^2 - t\epsilon)},$$

where the last step uses $t - \frac{1}{2} \ln(1 + 2t) \leq t^2$ for $t \geq 0$, proved in the same way as (1). Taking $t = \frac{\epsilon}{2}$ gives $\mathbb{P}(Z \leq 1 - \epsilon) \leq e^{-\frac{m\epsilon^2}{4}} \leq e^{-\frac{m\epsilon^2}{8}}.$

Conclusion. By the union bound,

$$\mathbb{P}(|Z - 1| > \epsilon) \leq \mathbb{P}(Z \geq 1 + \epsilon) + \mathbb{P}(Z \leq 1 - \epsilon) \leq 2e^{-\frac{m\epsilon^2}{8}}. \quad \square$$

By homogeneity, Proposition 3.2 extends immediately from unit vectors to arbitrary vectors: if $x \neq 0$, then applying the proposition to the unit vector $\frac{x}{\|x\|}$ and using linearity of M gives

$$\mathbb{P}\left((1 - \epsilon)\|x\|^2 \leq \|Mx\|^2 \leq (1 + \epsilon)\|x\|^2\right) \geq 1 - 2e^{-\frac{m\epsilon^2}{8}}, \quad (2)$$

and for $x = 0$ the event holds trivially.

4 From One Vector to Finitely Many

We now upgrade the single-vector estimate (2) to a simultaneous guarantee for a finite family of vectors.

Proposition 4.1. *Let $x_1, \dots, x_r \in \mathbb{R}^n$ and suppose $m > \frac{8 \ln(2r)}{\epsilon^2}$. Then with positive probability,*

$$(1 - \epsilon)\|x_i\|^2 \leq \|Mx_i\|^2 \leq (1 + \epsilon)\|x_i\|^2 \quad \text{simultaneously for all } i \in [r].$$

In particular, there exists $\omega \in \Omega$ such that the deterministic matrix $M(\omega)$ satisfies all r inequalities.

Proof. Let B_i denote the event that the inequality holds for x_i . By (2), $\mathbb{P}(\overline{B_i}) \leq 2e^{-\frac{m\epsilon^2}{8}}$ for each i . Hence, by the union bound and the assumption on m ,

$$\mathbb{P}\left(\bigcap_{i=1}^r \overline{B_i}\right) = \mathbb{P}\left(\bigcup_{i=1}^r \overline{B_i}\right) \leq \sum_{i=1}^r \mathbb{P}(\overline{B_i}) \leq 2r e^{-\frac{m\epsilon^2}{8}} < 2r e^{-\ln(2r)} = 1.$$

Therefore $\mathbb{P}(\bigcap_{i=1}^r B_i) > 0$, which proves the claim. $\square$

5 Proof of the Lemma

Proof of Lemma 1.1. Apply Proposition 4.1 to the difference vectors

$$\{u - v : u, v \in P, u \neq v\},$$

of which there are $r = \binom{N}{2} \leq \frac{N^2}{2}$. Then $2r \leq N^2$, so

$$\frac{8 \ln(2r)}{\epsilon^2} \leq \frac{16 \ln N}{\epsilon^2} < m,$$

and Proposition 4.1 produces an $\omega \in \Omega$ such that $M = M(\omega)$ satisfies

$$(1 - \epsilon)\|u - v\|^2 \leq \|M(u - v)\|^2 \leq (1 + \epsilon)\|u - v\|^2$$

for all $u, v \in P$ with $u \neq v$ (the case $u = v$ being trivial). Since M is linear, $M(u - v) = Mu - Mv$, which completes the proof. $\square$

Remark. The proof is non-constructive only in a mild sense: it shows that a random Gaussian matrix works with probability at least $1 - 2r e^{-\frac{m\epsilon^2}{8}}$, which can be made arbitrarily close to 1 by increasing m slightly. In practice, one simply samples M at random.

6 AI Usage

This note is polished using Fable 5, which makes the note clearer and easier to follow than the original version that I wrote as a non-native speaker of both English and French but have spent a large portion of my lifetime learning both of them.

References

- [1] W. B. Johnson and J. Lindenstrauss, *Extensions of Lipschitz mappings into a Hilbert space*, Contemporary Mathematics **26** (1984), 189–206.
- [2] Wikipedia contributors, *Johnson–Lindenstrauss lemma*, Wikipedia, The Free Encyclopedia. https://en.wikipedia.org/wiki/Johnson-Lindenstrauss_lemma
- [3] Concours BCE, *Mathématiques appliquées, épreuve ESSEC II*, annales 2026. <https://www.concours-bce.com/annales/mathematiques-appliquees-essechec-2>